Graphs of parabolas

1 Consider the equation $y = x^2$.

a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

b Use the table of values to sketch the graph of the function on a number plane.

c Are the y-values ever negative?

d Find the equation of the axis of symmetry.

e Find the minimum y-value.

f For every y-value greater than 0, how many corresponding x-values are there?

2 Consider the equation $y = -x^2$.

a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

b Use the table of values to sketch the graph of the function on a number plane.

c Are the y-values ever positive?

d Find the equation of the axis of symmetry.

e Find the maximum y-value.

3 For each of the following equations:

i Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

ii Sketch the graph of the function on a number plane.

a $y = 3x^2$ **b** $y = -2x^2$ **c** $y = 3x^2 + 3$ **d** $y = -2x^2 + 5$

e $y = 3x^2 - 3$ **f** $y = -2x^2 - 2$ **g** $y = \frac{1}{2}x^2$ **h** $y = -\frac{1}{2}x^2$

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Transformations of parabolas

4 For each of the following equations:

i Complete the table of values:

x	1	2	3	4	5
y					

ii Sketch the graph of the function on a number plane.

iii Find the minimum y-value.

iv Find the x-value that corresponds to the minimum y-value.

v Find the coordinates of the vertex.

a $y=(x-3)^2$ b $y=-(x-3)^2$ c $y=(x-1)^2$ d $y=(x-4)^2$

5 For each of the following equations:

i Complete the table of values:

x	-3	-2	-1	0	1
y					

ii Sketch the graph of the function on a number plane.

iii Find the minimum y-value.

iv Find the x-value that corresponds to the minimum y-value.

v Find the coordinates of the vertex.

a $y=(x+1)^2$ b $y=-(x+1)^2$ c $y=(x+2)^2$ d $y=-(x+3)^2$

6 For each of the following equations:

i State the coordinates of the vertex.

ii Solve for the equation of the axis of symmetry.

iii Sketch the graph of the function on a number plane.

iv Plot the axis of symmetry.

a $y=(x-2)^2$ b $y=(x+3)^2$ c $y=(x+5)^2$ d $y=-(x-5)^2$

7 Describe the transformation required to transform the parabola $y=-x^2$ into the parabola $y=-(x+4)^2$.

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8 State whether the following parabolas will be concave up or concave down:

a $y=(x-4)^2$ **b** $y=-(x+2)^2$ **c** $y=-(x-8)^2$ **d** $y=(x+7)^2$

9 For each of the following equations:

i Complete the table of values:

x	-2	-1	0	1	2
y					

ii Sketch the graph of the function on a number plane.

iii State the coordinates of the vertex.

a $y=2x^2$ **b** $y=3x^2$ **c** $y=-4x^2$ **d** $y=\frac{1}{2}x^2$

10 For each of the following equations:

i Solve for the equation of the axis of symmetry.

ii Sketch the graph of the function and its axis of symmetry on a number plane.

a $y=4x^2$ **b** $y=-3x^2$ **c** $y=\frac{1}{4}x^2$ **d** $y=-\frac{1}{2}x^2$

11 Describe the transformation required to transform the parabola $y=x^2$ into the parabola $y=5x^2$.

12 Rewrite the equation, $y=x^2$, after the following transformations take place:

a Vertically translated by 3 units.

b Vertically translated by -2 units.

c Horizontally translated by 5 units.

d Horizontally translated by -4 units.

e Vertically scaled by 2 units.

f Vertically scaled by -3 units.

g Vertically scaled by $\frac{1}{2}$ units.

h Vertically reflected about the x-axis.

13 Graph the equation, $y=x^2$, after the following transformations take place:

a Vertically translated by 4 units.

b Horizontally translated by -5 units.

c Vertically translated about the x axis.

d Vertically scaled by 2 units.

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Applications

14 William is training for a remote control plane aerobatics competition. He wants to fly the plane along the path of a parabola so he has chosen the equation:

$$y = 3x^2$$

where y is the height in metres of the plane from the ground, and x is the horizontal distance in metres of the plane from its starting point.

a Complete the table of values:

x (m)	0	1	2	3	4
y (m)					

b Sketch the shape of the path on a number plane.

c Find the lowest height of the plane.

d Find the x-value that corresponds to the minimum y-value.

e Find the coordinates of the vertex.

15 On Jupiter the equation, $d = 12.5t^2$, can be used to approximate the distance in metres, d, that an object falls in t seconds, if air resistance is ignored.

a Complete the table of values:

time (t)	0	2	4	6
distance (d)				

b Sketch the shape of the path on a number plane.

c Use the equation to determine the number of seconds, t, that it would take an object to fall 84.4 m. Round the value of t to the nearest second.

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